DSCI-564

Random Variable : $X: \Omega \to \mathbb{R}$

$\forall c \in \mathbb{R}$, $X^{-1}((-\infty, c]) \in F$

$= \{\omega \mid X(\omega) \in (-\infty, c]\} \in F$

$P: F \to [0,1]$

Borel σ-field $\mathcal{B}(\mathbb{R})$

Borel Set $B \in \mathcal{B}(\mathbb{R})$

$X^{-1}(B)$ is event

If $A \in F$, I_A is random variable

Probability Law

$P(X^{-1}(B)) = P(\{\omega \mid X(\omega) \in B\}) = P_X(B) = P(X \in B)$

$(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$ is a probability space

Independence of random variable

$\mathbb{P}(X \in B_1, Y \in B_2) = \mathbb{P}(X \in B_1)\mathbb{P}(Y \in B_2)$, $B_1, B_2 \in \mathcal{B}(\mathbb{R})$

Joint CDF : $F_{X_1X_2}(x_1, x_2) = P(X_1 \le x_1, X_2 \le x_2)$

if X_1, X_2 are independent $F_{X_1X_2}(x_1,x_2) = F_{X_1}(x_1)F_{X_2}(x_2)$

Properties of cdf

a) $P(X > x) = 1 - F_X(x)$

b) $P(x < X \le y) = F_X(y) - F_X(x)$

c) $P(X = x) = F_X(x) - \lim_{y \to x^-} F_X(y)$

iff $F_X(a)$ is continuous for all $a \in \mathbb{R}$

$P(X = a) = 0$, $\forall a \in \mathbb{R}$.

Cumulative Distribution Function (CDF) : $F_X: \mathbb{R} \to [0,1]$

$F_X(x) = P(X \le x)$

a) $\lim_{x \to -\infty} F_X(x) = 0$ $\lim_{x \to \infty} F_X(x) = 1$

b) $x < y \Rightarrow F_X(x) \le F_X(y)$; monotonic

c) $\lim_{x \to a^+} F_X(x) = F_X(a)$. right continuous

$B_n = \{\omega \mid X(\omega) \le a + \frac{1}{n}\}$. $A_n = (-\infty, a+\frac{1}{n}]$

$A_1 \supseteq A_2 \supseteq A_3 \cdots \Rightarrow B_1 \supseteq B_2 \supseteq B_3$

$\lim_{n \to \infty} B_n = \bigcap_{n \in \mathbb{N}} B_n = \{\omega \mid X(\omega) \le a\}$

$\lim_{n \to \infty} P(B_n) = \lim_{n \to \infty} F_X(a + \frac{1}{n})$

$= P(\lim_{n \to \infty} B_n) = P(\{\omega \mid X(\omega) \le a\}) = F_X(a)$

$\therefore F_X(a) = \lim_{x \to a^+} F_X(x)$

$$e^z = \sum_{l=0}^{\infty} \frac{z^l}{l!} \qquad \sum_{x=1}^{\infty} \frac{1}{x^2} = \frac{\pi^2}{6}$$

Combinations $P(A) = \frac{|A|}{|\Omega|}$, $\forall A \subseteq \Omega$

multiplication principal $|\Omega| = |A \times B| = |A||B|$

L ordered sampling with replacement

permutations $\frac{n!}{(n-k)!}$

L ordered sampling without replacement

Combinations $\frac{n!}{k!(n-k)!}$; binomial coefficient

L unordered sampling without replacement

$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, $\sum_{k=0}^{n} \binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}$

partitions $n_1 + n_2 + \cdots + n_r = n$ $\quad \frac{n!}{n_1! n_2! \cdots n_r!}$; multinomial coefficient

Discrete Random Variables

$X^{-1}(\Omega) = R_X$ is countable

probability mass function (PMF)

$P_X: \mathbb{R} \to [0,1]$, $P_X(x) = P(X = x)$

$P(X \in B) = P_X(B) = P(X \in B \cap R_X) = \sum_{x \in B \cap R_X} P(X = x)$

$F_X(x) = \sum_{y \le x} P_X(y)$, $\sum_{x \in \mathbb{R}} P_X(x) = 1$

a) Bernoulli $X \sim Ber(p)$

$P(X=1) = p$ $P(X=0) = (1-p)$

b) Discrete Uniform $X \sim dU(a,b)$

$P(X=x) = \frac{1}{b-a+1}$, $a \le x \le b$, $x \in \mathbb{Z}$

c) Binomial $X \sim Bin(n,p)$: n independent Bernoulli trials.

$P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$

d) Geometric

1) $X \sim Geo(p)$ $P_X(x) = p(1-p)^{x-1}$ $x \in \mathbb{N}$

2) $X \sim Geo(p)$ $P_X(x) = p(1-p)^x$ $x \in \mathbb{Z}^+$

e) Poisson $X \sim Pois(\lambda)$

$P_X(x) = \frac{e^{-\lambda}\lambda^x}{x!}$, $x \in \mathbb{Z}^+$

f) Negative Binomial $X \sim NB(k,p)$

$P_X(x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}$, $x \ge 1, x \ge k$

; k independent Geo trials.

$X_n \sim Bin(n, \frac{\lambda}{n})$ $X \sim Pois(\lambda)$

$\lim_{n \to \infty} P(X_n = x) = P(X = x)$

$X_N \sim HG(N, pN, n)$

$X \sim Bin(n, p)$

$\lim_{N \to \infty} P(X_N = x) = P(X = x)$

g) Power Law Distribution

i) Zipf-Mandelbrot $P_X(x) = \frac{C}{(x+q)^\alpha}$, $x = 1, 2, \cdots N$

ii) Hurwitz-Zeta $P_X(x) = \frac{C}{(x+q)^\alpha}$, $x = 1, 2, \cdots$

iii) Zeta $P_X(x) = \frac{C}{x^\alpha}$, $x = 1, 2, \cdots$. $\alpha > 1$; $X \sim Zeta(\alpha)$

iv) Zipf $P_X(x) = \frac{C}{x^\alpha}$, $x = 1, 2, \cdots, N$: $X \sim Zipf(\alpha)$

v) another $P_X(x) = \frac{1}{x^\alpha} - \frac{1}{(x+1)^\alpha}$ $x = 1, \cdots, \alpha > 0$; $X \sim Pow(\alpha)$

h) Hyper-geometric $X \sim HG(N, K, n)$

$P_X(x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$ $x \in \mathbb{N}$

; number of success in n draws w/o replacement

Joint pmfs $P_{XY}: \mathbb{R}^2 \to [0,1]$

$P_{XY}(x,y) = P(X=x, Y=y)$

Marginal pmfs

$P_X(x) = \sum_y P_{XY}(x,y)$, $P_Y(y) = \sum_x P_{XY}(x,y)$

if X, Y are independent

$P_{XY}(x,y) = P_X(x)P_Y(y)$

Conditional pmfs

$P_{X|Y}(x|y) = \frac{P(X=x, Y=y)}{P(Y=y)}$

Law of total probability

$P_Y(y) = \sum_x P(Y=y|X=x)P(X=x)$

Substitution Law

$P(Z=z) = P(g(X,Y)=z) = \sum_x P(g(X,Y)=z|X=x)P(X=x)$

$= \sum_x P(g(x,Y)=z|X=x)P(X=x)$

Derived Distribution

$Y = g(X)$. $P(Y=y) = \sum_{\{x \mid g(x)=y\}} P_X(x)$

i.g) $Z = X+Y$. X, Y are independent

$P(Z=z) = \sum_x P(Z=z|X=x)P(X=x) = \sum_x P(x+Y=z|X=x)P(X=x)$

$= \sum_x P(Y=z-x)P(X=x)$

Max, Min Problems : $Y = \max(X_1, X_2, \cdots, X_n)$ independent identical distribution

$F_Y(y) = P(X_1 \le y, X_2 \le y, \cdots X_n \le y) = F_{X_1}(y)F_{X_2}(y) \cdots F_{X_n}(y)$

$= F_X(y)^n$

Moments of Discrete random variables

$\mathbb{E}[X] = \sum_x x P_X(x)$

Law of unconscious statistician (LOTUS)

$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_x g(x) P_X(x)$

$\mathbb{E}[I_B(X)] = \sum_x I_B(x) P_X(x) = \sum_{x \in B} P_X(x) = P_X(B)$

$\mathbb{E}[X^2] = \sum_x x^2 P_X(x)$: second moments of X

skewness $\mathbb{E}\left[\frac{\tilde{X}^3}{\sigma_X^3}\right]$ kurtosis $\mathbb{E}\left[\frac{\tilde{X}^4}{\sigma_X^4}\right]$

a) $X \sim Ber(p)$ $\mathbb{E}[X] = p$ $V[X] = p(1-p)$

b) $X \sim Bin(n,p)$ $\mathbb{E}[X] = np$ $V[X] = np(1-p)$

c) $X \sim Geo_1(p)$ $\mathbb{E}[X] = 1/p$ $V[X] = \frac{1-p}{p^2}$ $\mathbb{E}[X(X+1)]$

$X \sim Geo_0(p)$ $\mathbb{E}[X] = \frac{1}{p} - 1$

d) $X \sim Pois(\lambda)$ $\mathbb{E}[X] = \lambda$ $V[X] = \lambda$ $\mathbb{E}[X(X-1)]$

e) $X \sim Pow(\alpha)$ $\mathbb{E}[X] = \sum_{n=0}^{\infty} P(X>n) = \sum_{n=0}^{\infty} \frac{1}{n^\alpha}$ (finite if $\alpha > 1$)

Stein's Characterization

$\mathbb{E}[X f(X)] = \lambda \mathbb{E}[f(X+1)]$

Cauchy-Schwartz-Bunyakovsky inequality

$|\mathbb{E}[XY]| \le \sqrt{\mathbb{E}[X^2]}\sqrt{\mathbb{E}[Y^2]}$

$\langle X, Y \rangle = Cov(X,Y) = \mathbb{E}[\tilde{X}\tilde{Y}]$

$\|X\| = \sqrt{\langle X, X \rangle} = \sqrt{Cov(X,X)} = \sqrt{Var(X)}$

Properties of Expectation

a) if $X \ge 0$ a.s. $\Rightarrow \mathbb{E}[X] \ge 0$

b) if $X = c$ a.s. $\Rightarrow \mathbb{E}[X] = c$

c) $a, b \in \mathbb{R}$, $\mathbb{E}[aX+bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ (Linearity of expectation)

d) if $X^{\to}(\Omega) = \mathbb{Z}^{\ge 0}$

$\mathbb{E}[X] = \sum_{n \ge 0} P(X>n)$

e) if X, Y are independent $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$

central moments $\tilde{X} = X - \mathbb{E}[X]$

$\mathbb{E}[\tilde{X}^2] = \mathbb{E}[(X - \mathbb{E}[X])^2] = Var(X)$

$\sigma_X = \sqrt{Var(X)}$

a) if $\mathbb{E}[X] < \infty$

$V[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

b) $a \in \mathbb{R}$. $V[aX] = a^2 V[X]$

c) X_1, X_2 are independent

$V[X_1 + X_2] = V[X_1] + V[X_2]$

d) $V[X] = 0$ iff $X = c$ as

$\mathbb{E}[X(X+1)]$

Existence of $\mathbb{E}[X]$, $V[X]$

$\sum_{x \le 0} x P_X(x)$ $\sum_{x > 0} x P_X(x)$

$\mathbb{E}[X]$ not well defined — both infinite

well defined — infinite — only one of them infinite

— finite — both finite $\mathbb{E}[|X|]$ is finite

$\mathbb{E}[X^2]$ always well defined

$\mathbb{E}[X^2] < \infty$ squared integrable $\Rightarrow \mathbb{E}[|X|] < \infty$ absolute integrable

$V[X]$	$\mathbb{E}[X^2] < \infty$	$\mathbb{E}[\lvert X \rvert] < \infty$
not well defined		X
well defined infinite	X	O
finite	O	

$Cov(X,Y) = \mathbb{E}[\tilde{X}\tilde{Y}] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$

a) $Cov(X,X) = Var(X)$

b) $\forall a, b \in \mathbb{R}$, $Cov(X, aY + bZ) = aCov(X,Y) + bCov(X,Z)$

$Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2Cov(X_1, X_2)$

X, Y are independent $Cov(X,Y) = 0$

$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}}$

a) $-1 \le \rho(X,Y) \le 1$

b) $\rho(X,Y) = \pm 1$ $\tilde{Y} = \pm a\tilde{X}$

Conditional Expectation

$\mathbb{E}[Y|X=x] = \sum_y y P(Y=y|X=x)$

$\mathbb{E}[g(Y)|X=x] = \sum_y g(y) P_{Y|X}(y|x)$

Substitution Law $\mathbb{E}[g(X,Y)|X=x] = \mathbb{E}[g(x,Y)|X=x]$

Law of total probability

$\mathbb{E}[g(X,Y)] = \sum_x \mathbb{E}[g(X,Y)|X=x] P(X=x)$

$= \mathbb{E}[\mathbb{E}[g(X,Y)|X=x]]$

Conditional expectation as random variable

$\mathbb{E}[Y|X=x] \Rightarrow \mathbb{E}[Y|X]$

$P(Y \in B | X=x) = \mathbb{E}[I_B(Y)|X=x]$ i.i.d

Wald's Equality $\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[N]\mathbb{E}[X]$

if X, N are independent

$\boxed{\mathbb{E}[\mathbb{E}[X|Y,Z]|Y] = \mathbb{E}[X|Y]}$

Existence of conditional expectation

if Y is integrable, non-negative ($\mathbb{E}[|Y|] < \infty$)

$\mathbb{E}[|Y| | X=x]$ is finite

Properties of conditional expectation

a) $\mathbb{E}[a|X] = \sum_y a P_{Y|X}(y|x) = a$

b) $\mathbb{E}[aX + bY|Z] = a\mathbb{E}[X|Z] + b\mathbb{E}[Y|Z]$

c) $Y \ge 0$ a.s. $\mathbb{E}[Y|X] \ge 0$

d) X, Y are independent

$\mathbb{E}[Y|X] = \mathbb{E}[Y]$

e) Tower property

$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y]$

f) total expectation

$\mathbb{E}[g(Y)] = \mathbb{E}[\mathbb{E}[g(Y)|X]]$

g) $\mathbb{E}[Y g(X)|X] = g(X)\mathbb{E}[Y|X]$

h) $\mathbb{E}[g(X)|X] = g(X)$

i) $\mathbb{E}[Y|X, g(X)] = \mathbb{E}[Y|X]$

Conditional Moment

$\mathbb{E}[(Y - \mathbb{E}[Y|X])^r | X]$

$r=2$: $Var(Y|X)$

$V[X|Y] = \mathbb{E}[X^2|Y] - \mathbb{E}[X|Y]^2$

$Var(X) = \mathbb{E}[Var(X|Y)] + Var(\mathbb{E}[X|Y])$

$\mathbb{E}[(\mathbb{E}[Y|X] - Y) g(X)] = 0$

$\hat{Y} = \mathbb{E}[Y|X]$

$\langle \hat{Y} - Y, g(X) \rangle = 0$

$\perp$ orthogonal to all possible $g(X)$

$\mathbb{E}[(\mathbb{E}[Y|X] - Y)^2] \le \mathbb{E}[(g(X) - Y)^2]$